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INTERNATIONAL EDUCATION STATISTICS AND RESEARCH: STATUS AND PROBLEMS*

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Abstract — Although governments throughout the world invest heavily in education, they spend remarkably little to monitor and evaluate their investment. Management of most of the world's education systems is done without adequate information and analysis. The international education database is often unreliable, and leaves out crucial measures of quality, process and output. Education research does not receive the same priority as research on agriculture, health or the economy. These shortcomings are due in part to the limited professional capacity most countries have to gather education statistics and carry out education research, along with strong resistance by educators and policymakers to being evaluated. Improvements will depend heavily on greater commitment at the political level to measuring education outputs, and on the emergence of global leadership in promoting comprehensive, reliable assessments of education systems.

I. INTRODUCTION

Governments throughout the world are underinvesting in educational statistics and research. Most spend between 10 and 25 per cent of their budgets on education — an amount equal to over 5 per cent of world GNP (UNESCO, 1993, 104). They expect those investments to promote economic growth, social equity and national integration. Yet little of that money is used to analyze educational systems. Government statistics on education are uniformly narrow, seldom timely and often unreliable. Most countries lack systematic measures of educational output. Cross-national educational assessments are rare. Few governments allocate significant resources to educational research. As a result, the world has only limited information with which to monitor and evaluate one of its major investments.

Just as remarkably, deficiencies in the world's capacity to monitor and evaluate educational systems have attracted little attention. There is no vigorous, global debate over the reliability of educational statistics, nor over what is learned in schools. Most existing concern comes from a widely scattered group of academics, and from middle-level national and international civil servants. National political leaders have focused, with a few exceptions, only on expanding enrollments and have ignored educational performance. No major global organization has taken up the issues of educational statistics, assessment and research, and sought to make them part of what constitutes good governance in a modern state. The problems afflicting the international statistical database on education, and our inability to assess properly educational outcomes, are not yet on the global agenda.

II. PRINCIPAL PROBLEMS

Existing statistics are unreliable

The existing international education database is far too often unreliable. To be sure, data on education are better overall than they were in the 1940s and 1950s. Then, approximately 50 countries regularly reported information on their education systems, and much of that was not readily comparable cross-nationally. Today some 175 countries regularly provide at least minimal information within the framework of the International Standard Classification of Education (ISCED). Data collected by the advanced industrial countries have expanded over the past several decades, and often exceed that requested by UNESCO. Several develop-

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ing countries have significantly improved the quality of their education statistics.

But quality, even where improving, is still too low overall. Many of the world's poorest countries have made little progress in producing reliable data on their educational systems. Some have moved backward, and today do a poorer job of collecting data than they did during the 1960s and 1970s. Some twenty to thirty countries suffer disastrous problems in generating reliable education statistics. Another fifty or so report data that have significant gaps and weaknesses.

Overall, the problem is with the quality of the data reported rather than with a failure to report. Only a few countries report little or no data. Typically such lapses occur when broader political strife renders impossible the normal functions of government. Occasionally they occur in countries that, although not experiencing political turmoil, possess extraordinarily inept state bureaucracies. More commonly, however, countries report data based on collection procedures that are fundamentally flawed, or on definitions that are either non-standard or poorly understood. It is on the ground, where raw statistics are gathered and processed, that the reliability of existing data is most commonly compromised (Chapman and Boothroyd, 1988). The principal problems afflicting global education statistics are at the national, rather than the international, level.

Examples, unfortunately, are easy to find. Education statistics from at least five of the world's nine largest countries (Nigeria, Brazil, Pakistan, Egypt and Bangladesh) are judged by most experts to have serious deficiencies (UNESCO staff, personal interviews, 12–13 November 1992). A number of the smaller Caribbean countries report only limited information on their educational systems (UNESCO, 1991a,c). Brazilian education statistics reported in 1982 that 5.1 million new students had enrolled in the first year of primary school, despite the fact that only 3 million children were in the relevant age cohort (Costa Ribeiro, 1990). In Uganda, the Ministries of Education and Finance cannot agree on how many primary teachers the country has — one (Finance) counts some 85,000 and the other (Education) some 140,000 (World Bank staff member, telephone conversation, 19 November 1992). Many countries, as a UNESCO official recently noted, "do not know what they spend on education." In Syria recently some 50 per cent more students graduated from high school than had been projected, causing serious problems for the country's higher education system. In the mid-1980s, Somalia reported nearly twice as many teachers as were later determined to be present in schools. Generally, the poorest, worst-educated countries also have the least reliable education statistics. But countries as diverse as the Congo, Jordan and Argentina consistently report unreliable information on their systems of education (UNESCO staff, personal interviews, 12–13 November 1992).

Problems are particularly severe in sub-Saharan Africa, where several decades of political turmoil and economic decline have seriously eroded an already weak statistical base. In many of those countries, the minimum set of trained people, institutions, funds, incentives and political will necessary to collect accurate education statistics is not in place. Specialists speak of the "collapse of the most basic information on education" in some of them (International Institute for Educational Planning staff member, personal interview, 13 November 1992). A process of state deterioration and deinstitutionalization is underway in Africa that profoundly undercuts the ability of those countries to collect and report reliable education statistics.

The countries of Central and Eastern Europe, and the former Soviet Union, also present a special challenge. For many years their education statistics were unreliable because of political pressures to demonstrate the success of Socialism. Now most of the old channels have been abolished, and new channels for collecting and reporting information are being set up. But the new systems are being developed within the context of serious economic hardship, and are not yet functioning well. Whether they will manage to produce reliable education statistics remains to be seen.

Worldwide, some of the most consistently unreliable data are on literacy. UNESCO's 1958 definition of an illiterate — someone "who cannot with understanding both read and write a short simple statement on his everyday life" — has been operationalized in different ways in different countries (UNESCO, 1990b). Many governments draw their literacy data from the census, based on asking respondents whether they can read and write. Some simply equate literacy with the proportion of the population

that has completed a certain minimum number of years of formal education. UNESCO considers four years of primary schooling as the cut-off point for such estimates. But at least one large industrial country recently used completion of the full six-year primary cycle as its cut-off point. Often UNESCO does not know how literacy data were collected (UNESCO, 1992b, 4).

Moreover, the concept of literacy is itself changing. Many countries emphasize "functional literacy," which comprises the minimum reading and writing skills necessary to function effectively in modern society. Functional illiteracy is of particular concern to advanced industrial countries, where it is widespread despite low levels of formal illiteracy. It remains difficult to measure, however, because no internationally accepted definition yet exists (UNESCO, 1993, 78–79). Traditional literacy statistics continue to treat literacy as if it were a black or white phenomenon, and fail to capture the often more important presence of functional illiteracy.

Educational expenditure data are also particularly unreliable, even in the advanced industrial countries. A recent comparison of IMF, UNESCO and OECD data revealed "substantial discrepancies" in figures on public expenditures. For example, estimates by the three institutions of the proportion of total public expenditure that Canada allocated to education in 1988 range from 11.9 per cent to 15.9 per cent. OECD figures indicated that the proportion of total public expenditure going to education in Austria in 1988 was nearly half again as great as did UNESCO's figures. These differences appear among expenditure data in developing countries as well. IMF figures suggest that India spent nearly 20 per cent less on education in 1987 than do UNESCO figures (UNESCO, 1992c).

Discrepancies in estimates of education expenditures derive in part from differences in data collection procedures. UNESCO gives governments little guidance on how to calculate educational expenditures, while the IMF provides a detailed manual. OECD obtains its public education expenditure data from the same source as UNESCO, but uses a different source for its data on total public expenditure. In most countries, moreover, IMF data is collected from a different branch of the government than is UNESCO data. The three institutions have not begun discussing ways to harmonize their figures, although Canada is undertaking its own special study of the problem.

More generally, educational expenditure data are not structured in ways that make them useful to decisionmakers. Some ministers of education have only the education budget to work with, because expenditure data reside at the ministry of finance. The minister may therefore know what is budgeted for teachers' salaries, but not what is spent. Similarly, systems of national accounts often are not classified by functions. Thus expenditures on education by the Ministry of Agriculture (on agricultural schools, for example) or the Ministry of Labor (on short-term vocational training) may not be readily available to the education ministry. And aggregate education expenditures may never be available in a single administrative location.

The central international education database resides, of course, at UNESCO/Paris. UNESCO's precursor, the International Bureau of Education, first began reporting on national trends in education in 1933, via the International Yearbook of Education. UNESCO carried out its first questionnaire-based survey in 1950, covering 57 member states. The process has developed steadily since then, and underwent a major qualitative change in 1978, when the newly adopted International Standard Classification of Education (ISCED) was made the official basis for the national and international educational statistics (UNESCO, 1992a). Governments now report education statistics to UNESCO through the regular annual questionnaires it sends them. UNESCO then compiles and publishes the data in its Statistical Yearbook, and in a variety of occasional publications. Today, UNESCO is the world's premier source of international education statistics. Its database has grown from 57 countries in 1950 to virtually every country in the world. It inaugurated, in 1991, a new, periodic publication — the *World Education Report* — that uses UNESCO statistics to develop international education indicators, and to analyze the state of education worldwide. Yet experts at UNESCO estimate that data from perhaps 70 countries — slightly less than half of UNESCO's member states — suffer serious reliability problems (UNESCO staff, personal interviews, 12–13 November 1992).

The statistical regime is too narrow

A second problem is that the information going into the international database on education is too narrow. The statistics that countries regularly gather, even when reliable, fail to cover some of the most fundamental aspects of education. Most resources have been dedicated to a single approach to monitoring educational systems: counting the inputs. Governments have concentrated almost entirely on recording the number of teachers, pupils and buildings in the formal system, and the size of public educational expenditures. They have given virtually no attention to documenting how schools function or what students learn. Therefore much of the data that could help determine where scarce educational resources ought to be allocated, or how effectively they are being used, are simply not on hand.

The narrowness inherent in this approach is both methodological and conceptual. Methodologically, governments have been dominated by what specialists often call a "head-counting" mentality that has led them to monitor educational progress through what is essentially a census. UNESCO asks them to supply a "full count" of pupils, teachers, classrooms and budgetary expenditures (UNESCO, 1992a). They attempt to establish a systemwide capacity to generate that count annually.

Relying almost exclusively on census-type head counts, however, has serious drawbacks. Head counts are expensive. They require stable institutions, considerable expertise and the ability to enjoin the entire educational system regularly to produce uniform statistics — resources that many countries lack.

A majority of governments, for example, fail to report the age data that UNESCO requests, making it impossible to calculate net enrollment ratios and admission rates. In-kind expenditures, such as help in building schools — often important in developing countries — are almost never reported. Only about half the OECD countries can report private expenditures on education by level (OECD, 1992). Most countries report nothing at all on private education expenditures, and UNESCO's Statistical Yearbook leaves them out completely. That omission is important. For example, there exist virtually no systematic data on the private costs of the "dual" system of vocational training that is in place in Germany, Austria and Switzerland, even though many argue that it is a model for other countries. The existing education statistics database is particularly ill-suited to advising countries how much they are investing in education and training.

Moreover, the current system of educational censuses fails to count many important inputs. Systematic global data on the training possessed by teachers at the various levels of education are unavailable. Information on the family backgrounds of students — key to identifying and addressing the special problems of migrants — is not generally collected. Nor is information collected on the availability of texts and other educational materials in the classroom. Many statistics cannot be disaggregated into such simple but important geographical classifications as rural and urban.

Major categories of education, such as non-formal and adult education, carried on outside the formal system are not included in the official statistics reported to UNESCO. Apprenticeships, short courses and part-time programs that occur outside the formal system are likely to be left out. Private educational facilities in general, including some religious schools, are not well covered. The information available on education worldwide refers almost exclusively to the formal education system.

The counting mentality that is inherent in traditional approaches to national education statistics has, in addition, focused attention on the most elementary forms of information about education, and worked against the development and use of more sophisticated measures. Policymakers are presented with basic statistics rather than with carefully developed indicators — for example, with enrollments rather than admission rates. Thus they do not learn the potential of such statistics, and may not allocate funds for them. The dominant approach, in the words of UNESCO specialists, is "collect and file" rather than "understand and use" (UNESCO staff member, personal interview, 12 November 1992). That emphasis does not encourage a more dynamic, instrumental approach to monitoring education systems among educators. It reinforces traditional tendencies to consider education in terms of stated policies rather than in terms of observed results. The government's analytic capacity, if present at all, is often located someplace other than the Ministry of Education.

The traditional dominance of a counting mentality has worked against developing agree-

ment regarding the definition of basic indicators, or even on whether a single standard is useful or possible. Unlike economics and public health, education does not enjoy systematic, internationally agreed-upon sets of indicators to aid program evaluation and policymaking. The functional equivalents of gross domestic product, open unemployment, morbidity and malnutrition do not exist for education. There are no internationally accepted indicators of educational quality or output. OECD has been unable to devise an indicator on teachers because member countries have no common definition of them. Data often include teachers who perform administrative or other duties, and mix indiscriminately full-time and part-time personnel (OECD staff member, personal interview, 12 November 1992). Most countries tend to leave out data on the private costs of education when making unit cost calculations, and to use only public recurrent costs (salaries). There has been, as a recent OECD publication observed, "little sophistication in using statistics as indicators" (OECD, 1992). Instead, education relies almost exclusively on elementary statistics.

Efforts at reform are beginning to appear. OECD's International Education Indictors Project (INES) is one such pioneering initiative, and has just published *Education at a Glance* — a "first set of international indicators" for its member states (OECD, 1992). UNESCO's *World Education Report 1991* (and the subsequent 1993 volume) represent an important effort to develop "the first set of world education indicators," despite the conceptual and quality limitations inherent in UNESCO's statistical database (UNESCO, 1992b). But these are very much first steps. And they are particularly deficient in measures of educational process, quality and output. There is as yet no robust, global debate on the development of educational indicators, nor general agreement that they are needed.

The traditional reliance on census-type counts is also ill-suited to monitoring some important aspects of education. Census-type counts are not good at measuring educational quality, which can probably be assessed more efficiently through sampling surveys. Nor are they likely to do a good job of capturing the educational process, which may be better measured through a wider range of quantitative and qualitative research. Just as governments use household surveys and elaborate indicators to complement their information on the economy, they might move beyond census-type counts and basic statistics to understand and assess their education systems. Yet sample surveys, indicators and modern social science research are conspicuously absent from the approaches most countries use to monitor their educational systems.

Conceptually, governments have focused almost exclusively on educational inputs, to the detriment of process and output measures. Most of the fundamental data included in the international database on education — classrooms, expenditures, teachers and pupils — are about inputs. UNESCO asks governments to report annually how many students they enroll at each level, but not how much those students are taught or learn. It requests information on the number of classrooms a country has, but not on what takes place within them. We systematically collect imperfect data worldwide on the costs of education, but we collect no such data on the benefits. The existing regime of education statistics, in the words of one specialist, is "fascinated with inputs" (UNESCO staff member, personal interview, 13 November 1992).

There are, for example, almost no measures of educational quality in the UNESCO database. We can approach the question of quality only by looking at statistics on repetition, dropouts and pupil–teacher ratios — all of them input-based measures, and at best uncertain proxies for educational quality. Nor do we have global information on educational processes, such as curriculum, language of instruction, teaching materials, number of hours spent in classrooms, or teaching methods.

Similarly, there is almost no systematic, global information on educational output. UNESCO's statistics include only one direct output measure: adult illiteracy. We can approach output somewhat less directly by looking at the number of children completing each year or level of the educational system. But we lack data that are richer, more qualitative, and that address directly the question of what schools produce.

Particularly conspicuous is the absence of educational achievement data. Most of the existing information lies outside the UNESCO system, and consists of national tests administered in perhaps 40 or 50 countries, and often

not comparable cross-nationally. By far the best source of internationally comparable data on educational outcomes is the International Association for the Evaluation of Educational Achievement (IEA). Over 30 years, IEA has generated a variety of cross-national educational achievement studies that have become global benchmarks, both for their methodology and for their results (Hayes, 1992; Elley, 1992 and Husén, 1967). But IEA studies so far have concentrated heavily on advanced industrial countries. They also are not periodic, and are driven more by the concerns of research than by a policy agenda. Thus although IEA studies may be helpful to those policymakers who read them, they do not provide the systematic, comprehensive view that policymakers need.

More generally, the idea of regularly assessing the performance of educational systems through timely, empirical data has not yet taken root. Very few countries have their own systems for doing so, and no global system has begun to take shape. UNESCO, for example, has had limited involvement in achievement-testing over the years. As a recent UNICEF document noted: "In most countries today there is no objective measurement to monitor the real progress made in achieving basic education objectives, essential in light of the Jontiem conference" (UNICEF, 1990a). These problems often stand in sharp contrast to economic, demographic and health statistics, which tend generally to be more comprehensive and reliable than those available on education.

Governments have focused their resources on just one, relatively narrow, approach to monitoring educational systems — a census-type count — and to a single type of data — inputs. That regime requires institutions and capacities that are beyond the reach of many countries, and leaves out many elements that are crucial to evaluating educational performance.

Research is not widely available, or widely used

A third problem is that many, if not most, countries fail to complement statistics with research in their efforts to monitor educational systems. To be sure, educational research has grown significantly worldwide over the past two decades. Researchers, institutions, publications and training programs have increased dramatically. A series of subregional research networks have been established, and are becoming more active. Cross-national research is more common, and more sophisticated, than it was twenty years ago.

But those advances are nonetheless limited in scope, and trail significantly behind progress made in such fields as agriculture and health. The Consultative Group on International Agriculture Research (CGIAR), for example, raises more than $300 million dollars annually just for its 13 international agricultural research centers (CGIAR, 1992, 33). Those funds are complemented by over $8 billion dollars each year in public spending on agricultural research at the national level (Pardey *et al.*, 1991). In health, approximately $35 billion dollars is spent each year on research worldwide (Commission on Health Research for Development, 1990). By contrast, reliable estimates on support for educational research worldwide are not even available, because no institution has been delegated and funded to gather them. Nothing similar to the CGIAR system exists for education, however, and there is no significant global initiative to promote research on education.[1]

Educational research remains underfunded and underappreciated worldwide. Even in the United States, federal funding for educational research between 1980 and 1987 declined by one-third in real terms, while federal support for research in all areas grew by one-quarter (Wise, 1991).[2] Many countries carry out no educational research at all. Most carry out significantly less research on education than they do on the economy, or on the public health system. Too much of what is carried out is weak. Governments worldwide have not made educational research a high priority.

In the absence of good research, education often escapes the rigorous, empirical analysis that is applied more commonly to the economy or to the public health system. Countries tend to analyze their investments in education less, and less carefully, than they do many of their other investments. And even when they identify problems facing students and schools, they often fail to understand their causes. We know considerably more about costs, processes, output and demand associated with industrial production than we do about their counterparts in education. We have not studied the determinants of learning and the

impact of teaching the way we have studied the prevention and treatment of disease.

Education policy therefore is too often based on imprecise data, guessing or purely political considerations. It is too seldom informed by efforts to confirm, explain and evaluate educational outcomes. In consequence, policy declarations tend to become the center of attention rather than policy results. For much of the world, educational research has not yet become one of the tools policymakers use to guide their decisions.

Demand is growing more rapidly than supply

Education statistics and research have not only lagged behind education's importance to society, but also behind society's interest in them. Although the effective demand for better educational assessment remains incipient, it has increased significantly over the past five years. The advance of democracy, new approaches to economic policy, greater attention to the performance and accountability of governments, and greater access to information are generating a new and more sophisticated understanding of education and its uses. Whereas enrollments were a major concern throughout the 1970s, today the major emphasis is on quality. Increasingly, parents are concentrated about the quality of the education their children receive, rather than the quantity. Employers compelled to compete in an international economy take a sharper interest in the skills their employees possess. Politicians facing growing demands for accountability are looking more carefully at the education budget. Journalists aware of educational reforms in other countries ask more questions about schools in their own. Each is asking harder questions about how schools work and what they produce.

The demand for knowledge about educational systems will continue to increase over the next decade or more. It will focus progressively more on questions of quality and outcomes. Most of the data necessary to answer those questions is not systematically collected today. This means that the users of education — students, parents, and employers — will increasingly find their questions unanswered. They will be less willing than in the past to accept that absence of information, and more willing to withhold political support for education authorities. It means that political leaders will decide on education policy without information that might help them make good decisions. Thereby, education may not be as appropriate, as productive or as efficient as it ought to be in a world where good education is increasingly fundamental to economic welfare and social harmony.

III. PRINCIPAL CAUSES

Education is complex

Some portion of these problems owes, of course, to the fact that education is a complex phenomenon. Learning begins at birth, and continues throughout a lifetime. It takes place not only in schools, but also at home, at work, and on the street. And it resides in the mind, which is relatively resistant to direct observation and precise analysis. There is also considerable disagreement over what kinds of learning deserve attention in the context of education, and whether they can be measured.

That complexity is reflected in the difficulties countries experience when they record, classify and report education data to UNESCO. Specialists tend to agree that the International Standard Classification of Education (ISCED), developed by UNESCO in 1978, is outdated, culturally bound and sometimes impossible to use. ISCED's concepts and definitions are especially inappropriate for vocational and technical education, but need revision more generally to reflect the major changes that have occurred over the past two decades at several education levels.

Classification problems are just one example of how education's complexity makes more difficult efforts to measure the education process. But education's complex character does not mean that we cannot measure and evaluate it. It simply means that we must measure it carefully, in diverse ways and taking into account its multifaceted nature. Most specialists agree that extensive data useful for monitoring and evaluating education can, technically, be developed. They also agree that so far most such data has not been developed. In most countries, the constraints on measurement that education's complexity poses have not been a problem because so little measurement has been attempted.

Money is not always available

Another explanation for the world's limited progress in monitoring and evaluating education is money. Most countries are poor. Some, particularly in Africa, face extraordinarily severe economic crises. In times of economic hardship, governments are more likely to cut statistics than other government functions. More commonly, governments have been hard-pressed to keep up with rapidly expanding enrollments — exacerbated by the demographic transition — and have ignored or given low priority to monitoring their educational systems. Funds that might have gone to statistics, analysis and information have gone instead to build new schools and pay teachers' salaries.

But lack of money is only one explanation, and in most cases it is probably not the most important. All nations have some money. Virtually all of them decide to spend relatively small amounts on monitoring their educational systems. The fact is that most countries do not assign high priority to statistics and research. They spend the money they have on other items. Better monitoring of educational systems is less a problem of funding than of priorities.

National capacity is often weak or absent

A more fundamental explanation for most countries' failure to properly monitor and evaluate educational systems is the lack of a national capacity to do so. UNESCO's statistics, after all, are only as good as the data governments report. It has no independent ability to gather such data and must, as one internal document notes, "maintain a broad consensus among [governments] as to the amount and type of data that can be requested" (UNESCO, 1992a, 5). National capacities to collect data on educational systems constitute, therefore, a major constraint.

Unfortunately, many, and perhaps most, countries lack a professional capacity to gather good education statistics. One good proxy for a sophisticated data collection mechanism is the ability to report data on the ages of pupils/students, which are necessary for calculating net enrollment ratios and school life expectancies (UNESCO, 1993). Only some 60 of UNESCO's 175 member states consistently report such data. Only about half of the remaining countries are able to provide the other data UNESCO requests within reasonable levels of accuracy. The remainder, some 60 countries, find it very difficult to do so. The large developing countries are particularly important in this respect, since they account for such a great proportion of the world's children in school. At least five of the nine largest developing countries encounter serious difficulties in collecting complete and accurate education statistics. Many countries simply cannot now meet UNESCO's global statistical requests (UNESCO staff, personal interviews, 12–13 November 1992).

In some countries, these problems stem from a broader inability to govern, brought on by civil strife or economic disaster. But often the capacity problem is more specific: governments have not allocated the funds necessary to establish, equip and staff a proper education statistics office. The people on hand may not have been trained for data collection. They may lack specific expertise in computerizing data. They may change jobs frequently, providing no continuity from one year to the next. They may not have enough money. There may be no effective demand, from the media, policymakers or parents, for better data on education. There may be no incentives to produce it.

In some countries education statistics are given less priority than the other statistics that governments routinely collect. Governments often "delegate" the gathering of education statistics to the ministry of education with little funding or supervision. Thus even when governments collect relatively reliable statistics in other areas, they may fall short in education. Often the system for collecting education statistics is totally separate from the country's other statistical systems, and the central bureaucracy makes no effort to develop expertise in the education area.

Where capacity is a problem, additional money spent on gathering education statistics may be wasted. Units that lack training, equipment and organization are unlikely to spend properly the resources already available. Thus efforts to improve the quality of data reported by such countries must address first the capacity question. It is difficult for UNESCO, or any other international body, to engage officials in discussions about classification and definitions when those people are more concerned with securing computers or staff. Generating better education statistics, for many countries, depends directly on capacity-building.

Educational research capacity faces even more serious problems. Its status in developing countries, according to a 1991 review organized jointly by UNESCO's International Institute for Educational Planning (IIEP) and the Institute of International Education (IIE) of Stockholm University, "has been uncertain and it has functioned in a milieu of insecurity and isolation" (UNESCO and ILEP 1991b, 1). An assessment currently underway at UNESCO refers to "the poor capacities for, and the paucity of, indigenous evaluation and research in developing countries" (Chinapah, 1992). These deficiencies contrast with the areas of economics and health, where developing countries have made significant advances in establishing research capacity.

Capacity to carry out educational research is weak or non-existent in most developing countries. Properly trained people, effective incentives and appropriate institutions are often not in place. The research capacity that does exist tends to be fragile, dominated by a few key individuals, inadequately funded and unable to train its successors. And even when solid research capacity has been established, it often depends for funding almost entirely on foreign donors, or is inappropriately controlled by government authorities. In most developing countries it tends to have few effective links to policymakers. As a result, the capacity that is in place nationally produces research that is of mixed quality, and has little impact upon policy.

An assessment mentality is missing

Why is capacity to monitor and evaluate educational systems so weak worldwide? Why do governments fail to assign high priority to such activities? Why do parents, policymakers and the media seldom demand them?

Much of the answer derives from the widespread absence of an assessment mentality in education. The idea that education can and should be systematically assessed is not well established around the world. Governments, the general public and even educators may have their doubts, although often for different reasons.

Some believe that learning is too elusive to be measured, or if measured, too culturally specific to permit comparisons. For such people, education produces impacts that are so diverse, indirect, unpredictable and even indescribable, that to measure any one of them is to miss the point of what education is all about. Measurement, from that viewpoint, risks producing assessments that are seriously biased, because so many possible outcomes remain unmeasured. And international comparisons reflect, principally and inappropriately, the values and priorities of the countries that prepare the measuring instrument. Indeed, many who resist measurement in education do so for political as well as epistemological reasons. A significant group sees the quantitative research paradigm as being not only unrealistic, but also imperialistic — a self-serving construct being promoted by the advanced industrial countries. Although it is difficult to determine how widespread that view is, it is clearly consequential.

Another group, larger and more important, simply is not accustomed to measuring many aspects of education, and does not accord it priority. Attempts to improve the quality of data or to devise new kinds of assessment are perceived as a burden or a complication. Here the problem is partly ignorance and partly inertia. Because most aspects of education have for so long gone largely unassessed and unmeasured, it takes time to make the case for a new approach, and to establish new procedures. People must be made aware of the importance of assessment. A culture of measurement needs to be established.

This attitude is remarkably widespread, even in countries with substantial experience with educational assessment. OECD, for example, has only recently received comparable education indicators for 1988 from its member countries. It hopes to improve that record gradually, securing 1991 data by the end of 1993. Moreover, it took a year to convince OECD countries that early childhood education was a sufficiently important part of the educational system to warrant including it in the OECD indicators project (OECD staff, personal interview, 12 November 1992).

In developing countries the absence, in education, of a culture of measurement is probably the norm. UNESCO specialists talk of encountering "no awareness of, or belief in, the importance of statistics" among some government officials. They describe functionaries who make beautiful declarations regarding data collection, but subsequently carry out no real activity. They report the absence of a "climate" that fosters attention to statistics. The

mentality, according to one, is that you do not need statistics or research for policy. Another reports that "they don't really believe in it." Often the bureaucracy, standards, and traditions necessary to produce educational assessment do not exist. Not all countries or all government officials elicit such comments. But clearly there is, as yet another specialist put it, a "climate-building" problem (UNESCO staff, personal interviews, 12–13 November 1992). The benefits of educational assessment need to be made clear. Peer pressure needs to be generated. Governments must perceive that this is what good government is about.

Because statistics and assessment are not seen to be important, governments often pay little attention to the health and functioning of those units that collect data on the educational system. Thus, for example, investments in training often do not pay off because the trainees are subsequently not employed to do what they were trained to do. As one specialist observed, "You go back, and you find someone else" (UNESCO staff member, personal interview, 13 November 1992). Just as often, the incentives necessary to attract and maintain qualified people are not in place. When appropriate national structures — salaries, leadership, prestige and demand — are absent, external inputs, such as training, equipment and technical advice, are unlikely to make a significant difference.

Evaluation is politically unpopular

Yet another explanation for weakness in the worldwide capacity to assess and monitor educational systems is political resistance to being evaluated. This factor is difficult to analyze, because it is rarely expressed. Those motivated by a desire to avoid documenting the shortcomings of their educational systems are unlikely to say so, and may not even realize the source of their resistance. The fear of evaluation, however, is an important obstacle to better educational monitoring and evaluation.

Comprehensive, reliable assessments of educational systems may threaten important groups, including politicians, teachers, school administrators, government bureaucrats and policymakers. Evaluation, after all, involves judgment, and may bring change. Measuring efficiency and output may expose some to criticism; others may risk losing their jobs. It may uncover political patronage, inefficiency, incompetence and low standards. It may generate demands for people to learn new techniques and to work harder. It may energize the media and parents to pay more attention to how schools function. It is unlikely to support the status quo. Good analysis and a solid education database would be revolutionary in most countries. They would change the paradigm of educational management.

Most policymakers seek to avoid criticism, and many therefore perceive better education indicators as a threat. Few ministers of education enjoy having the deficiencies of their educational systems demonstrated through solid, empirical data. Few school administrators would want to publicize the fact that many of their students, after spending several years in classrooms, have learned very little. Few politicians are enthusiastic about the prospect of finding their country at the bottom of the list in international comparisons of educational achievement.

This attitude is common in all countries, rich or poor. The Ministry of Education in Mexico, for example, is reported to have education outcomes data that it refuses to publish because the results are uncomplimentary (IIEP staff member, personal interview, 12 November 1992). Chile, despite having been a pioneer among developing countries in educational reform, has been reluctant to make public the results by school of its annual primary learning assessments. OECD faced serious pressure recently from its member countries to publish the results from its education indicators project (INES) alphabetically, rather than in rank order. Rank-order data revealed that Sweden, for example, was not the best-educated country in Europe, and that the Netherlands was spending relatively large amounts on higher education, and relatively small amounts on compulsory education. That generated, according to one expert, a "tremendous resistance to ranking" among OECD members. Governments were worried about the effects such rankings might have on national policy, and on the images that policymakers had created (OECD staff member, personal interview, 12 November 1992).

Moreover, some argue that demonstrating the shortcomings of existing educational systems has no upside potential. "We already know things are bad," goes the argument, "so why evaluate?" The results will be un-

pleasant, may incur the wrath of the public, and are unlikely to increase prospects for improvement. Resources spent on assessment, according to this position, could be better spent on improving programs that we already know are defective.

This attitude makes countries particularly reluctant to compare themselves with others in terms of academic achievement. They see no positive benefit in doing so, and are acutely sensitive to the possibility that they will rank at or near the bottom in such comparisons. It is not surprising, for example, that the International Association for the Evaluation of Educational Achievement (IEA) generates relatively little interest among the poorer countries. Policymakers are not keen on the prospect of receiving bad news. Thus they are not generally committed to making educational comparisons based on solid, clear empirical data. And the very idea of cross-national comparisons of educational achievement therefore encounters either apathy or serious resistance. Comparability is not yet a legitimate concept in education.

This leads us to a more basic problem: the case for cross-national educational assessments has not been made effectively. Too many countries focus only on the negative aspects of such assessments, and ignore the potential benefits. Some part of this problem is technical. Many countries simply do not understand how cross-national assessments may help them improve their educational systems. There is a need to make that potential clear.

But achieving greater acceptance of cross-national educational assessments is not just a technical issue. It is also a question of political legitimacy. At present such assessments are heavily identified with institutions and researchers in the advanced industrial countries. They have little visibility and currency in the developing world. They have not been placed effectively upon the global agenda.

Accordingly, governments can easily ignore them. Pressure to participate in cross-national educational assessments comes principally from a few determined researchers or policymakers. Only in a few countries is public opinion beginning to demand more information on educational output, and thus playing a role as well. These efforts, however, are widely scattered, and they lack a global stamp of approval. There is no prominent, international consensus that such activities should be given serious consideration by all countries. The only two fora that could probably generate that political legitimacy — UNESCO and ECOSOC — have so far not chosen to take up the task.

Global leadership is missing

An important part of the problem, therefore, is that a supportive international climate for improving the world's capacity to monitor and evaluate educational systems does not yet exist. Conventional wisdom remains largely unchanged. National officials still do not see why they should invest time and effort in such activities. No comprehensive, high-profile effort to raise consciousness and change behavior around the world is underway. There is no strong international leadership for the cause of better educational assessment.

To be sure, several steps are being taken in the right direction. The OECD indicators project, for example, is a pioneering effort to develop education indicators, and marks a significant new direction for the advanced industrial countries that are participating in it. Similarly, the 1990 Jomtien *World Declaration on Education for All* urges countries to "improve and apply systems of assessing learning achievement" (UNESCO 1990c, 5). That call has prompted the UNICEF/UNESCO Joint Committee on Education to establish a project to develop "national capacities in monitoring basic education goals," which includes inviting a few countries to experiment with school-based sample surveys to measure learning achievement (UNESCO undated, 1). Although this project constitutes, in profile and funding, a minor part of UNESCO's activities, it is nonetheless a promising experiment. The World Bank has promoted the establishment of educational assessment systems in several developing countries, so far with only limited success (Horn *et al.*, 1991).

But most of the current initiatives to strengthen educational assessments reside substantially in the advanced industrial countries. IEA and INES have their roots there. The recent initiative to reform ISCED, as a UNESCO document notes, "originated in member states of the OECD and EEC" (UNESCO 1992b, 9). For the rest of the world (the UNESCO/UNICEF initiative notwithstanding), concern for better approaches to educational monitoring and evaluation are, at best, incipient.

Similarly, the IEA's ongoing difficulties in securing funding for its central operations, and for its projects in developing countries, underscore the absence of international leadership in promoting more and better educational achievement studies. The bulk of IEA's funding comes from European and North American sources, and underwrites studies in those countries. Funding for studies in developing countries is scarce and often precarious. There is little evidence that the pattern of demand is shifting.

Even among the advanced industrial countries, it is not clear that IEA's activities are highly valued. With the exception of the United States, Canada, the Scandinavians and the Netherlands (and, to a lesser extent, Japan) the advanced industrial countries have provided relatively modest support beyond funding their own country studies. Some governments have argued that they are structurally not able to give money to a non-governmental organization. Others have not understood the costs and value of international achievement studies. Together, they have not taken the initiative to put IEA on a sound financial footing, or to provide funds to expand activities significantly in developing countries. Educational achievement studies therefore remain largely the province of a group of highly committed researchers. No broadly credible international body has taken the lead in making such studies a prominent and regular part of educational assessment worldwide.

Another symptom of the tepid interest worldwide in improving educational monitoring and evaluation is the fact that UNESCO has not given such activities special priority. UNESCO is the world's chief source of information on education. Its database is the largest and most comprehensive in existence. But UNESCO has suffered severe economic difficulties over the past decade, and has had to cut back all of its operations. Significantly, it has trimmed resources allocated to its statistical function by roughly the same proportion as resources allocated to other activities. Arguably other parts of UNESCO's program are less essential to conduct through a worldwide intergovernmental organization. But UNESCO has not singled out education statistics, or other aspects of educational monitoring, for special protection from its budget cuts.

As a result, UNESCO's capacity to carry out its statistical function has declined. The organization's statistical unit has shrunk just like the rest of its branches, losing between 35 and 50 per cent of its staff over the past decade. Each regional office used to have a senior statistical officer; today none do. The post of senior professional responsible for statistics on educational finance has been abolished. Recently 34 African countries requested assistance in developing national educational statistics centers. Yet Paris has only two staff members who can provide such assistance, and they can help just six countries each year.

Thus although the unit still carries out its core function of statistical tabulation and storage, it has significantly less capacity to undertake such related activities as checking up on the reliability of statistics that governments report, promoting sound statistical practices, publishing partial analyses of its data and providing training in data collection and classification. At a time when demand for information and training from developing countries has been growing, the statistical unit has became a weaker organization.

UNESCO's response to these changes has been to treat its statistical function approximately the same as its other functions. The final report of the "Meeting of Experts on Education Indicators and the International Standard Classification of Education (ISCED)" that took place in Paris in June, 1992 urged that UNESCO "mobilize additional extrabudgetary resources for assistance to developing countries in strengthening their national education statistics systems" (UNESCO, 1992b, 12). The organization's capacity to mobilize new resources and to provide the complementary services could depend, however, on the priority it is willing to give to those functions within its current budget.

UNESCO's recent initiatives are encouraging, but remain limited. And they have not been matched by a concerted movement among the organization's member states to place education monitoring and evaluation high on its program agenda. Without that response, it will be difficult for UNESCO's Secretariat to move much further. Serious effort from a strong group of member states will be needed if the organization is to reassert leadership over one of its original mandates.

NOTES

1. UNESCO has sponsored several regional educational research networks in developing countries, with generally positive results. That initiative pales, however, when compared to similar efforts in agriculture and health.

2. Heyneman (1993) points out that Wise's figures imply that the U.S. government spends more than three times as much on health research in developing countries than it spends on education research on U.S. citizens.

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